

QuietKey Gate A Bridge Run · Pre-Registration

bridge-prereg-01 · Status: FROZEN. This document is committed to the repository before the first sheet is printed. Pass conditions below cannot be modified after printing begins; a change requires a new pre-registration and a new run. This applies the QK2-04 Gate A2 no-moving-goalposts rule to the bridge run itself.

1. Question under test

The spike measured the damage envelope on token sheets (dense 15-line layout, in-sheet T0 control). The two production-format locate pages were tested pristine and lightly worn only; damage has never been applied to a production page. The bridge run fills that cell: does the measured envelope transfer to the real Recovery Document format, where (a) the footer must first be FOUND among body text, (b) no control token exists, and (c) the reader runs exclusively in its T0-free production mode (sentinel gate plus confidence selection, per-line comb, global bank)?

2. Materials and fixed parameters

19 sheets S28-S46 plus 2 dim recaptures per bridge-log-01. One printer, one paper stock, one capture camera, all identical to the spike; the multi-printer matrix remains a separate later Gate A1 obligation. Reader operating point is FIXED at conf_floor 0.64, margin_floor 0.02; no re-tuning on bridge data under any outcome. Classifier bank: the spike global baseline bank (S01 plus S02), valid because printer and paper are unchanged. The committed spike reader files remain frozen as-run; bridge analysis is new code extending the locate-page path, producing a new dated report (bridge-verdict-01) and entering the repository under the same manifest and CI discipline. Reference envelope for comparison: T0-free 116 of 135 tokens; locate pages 2 of 2. Every reported decode success requires RS(83,49) plus AEAD verification against the T5 Vault Key.

3. Confirmatory pass conditions (gate the run)

B1, session validity: both pristine controls S28 and S46 decode. A control failure invalidates the session (print, light or focus fault) and requires a session reshoot; it is not a bridge failure. B2, benign transfer: all four benign-family maximum-severity sheets decode: scratch-3 (S39), scuff-3 (S40), fade-3 (S41), edge-3 (S42). These cells decoded 4 of 4 on token sheets; one production replicate each confirms transfer. B3, locator integrity: on every sheet, either the footer trio is correctly identified, or the failure is attributable by photo inspection to physical destruction or displacement of footer ink itself. A locator failure caused by body-text decoy ink on an intact footer fails this condition. B4, failure mode: every non-decoding token fails erasure-dominated and loudly. Zero silent end-to-end corruptions across the entire run: no decode output may exist that failed checksum or AEAD. B5, no parameter movement: thresholds, bank, and budget accounting are exactly as fixed in section 2. Converting a failure to a pass by any parameter change voids the run.

4. Rate-recording cells (reported, not gated)

Boundary treatments coffee-2 (S29-S30), crumple-2 (S31-S32), water-2 (S33-S34), water-3 (S35-S36): token-sheet decode rates were 2/4, 0/4, 2/4 and 2/4. Two replicates cannot estimate a rate; outcomes are recorded and compared against the spike cells, gated only through B3 and B4. Marginal cells crumple-1 (S37, spike 3/4) and fold-3 (S38, spike 3/4): decode expected; a failure is within the measured envelope, recorded, not gated.

5. Exploratory cells (predeclared)

E1 body ring (S43), E2 clip tear (S44), E3 span stain (S45), E4-E5 dim recaptures of S28 and S40. Outcomes are recorded; any failure produces a NAMED READER REQUIREMENT for the terminal implementation, never a bridge failure. Predeclared expectations: dim fails (the spike already placed single-frame dim capture below the envelope). The clip tear removes only URL-prefix camouflage characters, zero payload; if line-1 payload misindexes because run-start anchoring shifts, the named requirement is sentinel-anchored line-1 registration or whole-line-1 erasure fallback. The RS arithmetic makes that fallback legitimate by design: line 1 carries exactly 48 payload characters, 30 codeword bytes, inside the 34-byte erasure budget, so a token remains recoverable with its entire first line erased provided lines 2-3 are clean.

6. Outcome interpretation, fixed in advance

All of B1-B5 pass: the envelope transfers; proceed to freezing the Gate A1 acceptance thresholds on the combined spike-plus-bridge evidence, then the definitive multi-printer multi-camera matrix, with the Gate A2 salience pilot in parallel. Any of B1-B5 fails: named investigation; no threshold freeze until the cause is understood and either fixed in the reader (new pre-registered confirmation run) or accepted as a documented envelope boundary. Under no outcome are bridge results merged into spike result files; bridge-verdict-01 is a separate dated report and the spike artifacts remain frozen.